Data mining unit 3 classification

DATA MINING UNIT-3

CLASSIFICATION

1. Classification :- Basic Concepts
2. Decision tree & Decision Rules
3. Bayes classification methods.
4. Advance methods
5. Bayesian Belief Network
6. K-Nearest Neighbor (KNN) classifier
7. classification By Back propagation
8. Support Vector Machine.

CLASSIFICATION AND PREDICTION

Classification :- A classification technique is a Systematic Approach to build classification models from an input dataset; that is used to identify class label.

⇒ This input dataset is Known as Training dataset. In classification we Arrange new data with the help of Current or past data.

There are two Steps in classification

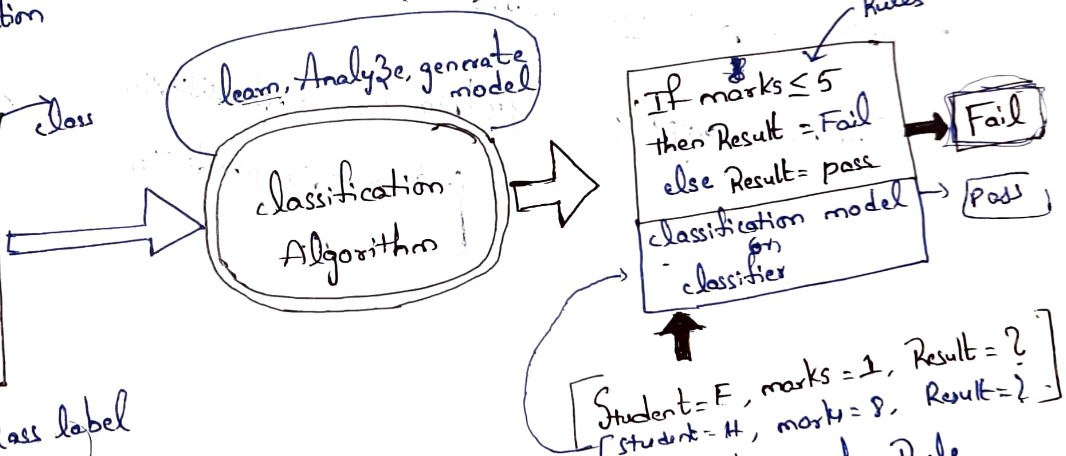
1. model Construction
2. model usage

Student	marks	Result
A	5	Fail
B	6	pass
C	2	Fail
D	7	pass
E	9	pass

Training data set

class

pass } class label
fail }



⇒ What ever data is given to this model, based on the Rule it will display class label.

⇒ Some of the classification Algorithms are Decision tree, Neural network, Bayesian network etc.

Prediction :-

⇒ Prediction is the process of identifying the missing or unavailable numerical data for a new object.

⇒ Algorithm which use training data set to derive a model, that model is predictor when the new data is given, this model should find a numerical output.

⇒ Unlike in classification, this model does not have the class label, this model predicts a continuous valued function or ordered value.

- Purchase Computer - yes/No ⟶ classification
- cost ⟶ prediction

Bayesian Belief Network

↳ The term Bayesian denotes "Statistical method" based on Baye's theorem

⇒ It provides a Simple way of applying Baye's theorem to Complex problems.

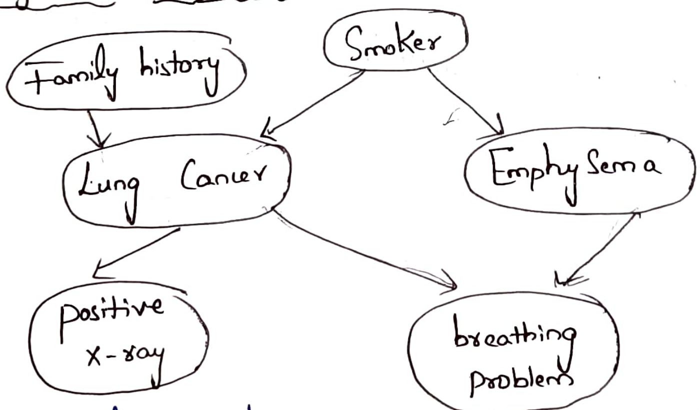
Definition:- Bayesian Belief Network is a probabilistic graphical model that represents a Set of Variables and their Conditional dependencies through a directed acyclic Graph (DAG).

there are two important phases in Bayesian Belief Network they are

1. Directed Acyclic Graph (DAG)
2. Conditional probability table (CPT)

1. Directed Acyclic Graph (DAG):

Example:



Attribute represents character of a person

⇒ Here each node of network is either random Variable or Attribute (Lung Cancer, Smoker)

Arrow mark Shows relationship between Variable or Attributes

② Conditional probability table (C.P.T):

Now we are finding probability of getting lung Cancer, lung Cancer depends on two Variables they are family history and Smoker. So, we Consider both family history and Smoker as parent Variables of lung Cancer. Here there is an example of Conditional probability table.

	FH, S	FH, ~S	~FH, S	~FH, ~S
LC	8	5	7	1
~LC	2	5	3	9

LC → Lung Cancer
~LC → without lung Cancer
FH → Family history
~FH → without family history
S → Smoker
~S → non-Smoker.

⇒ By using Conditional probability table we can represent Conditional probability of the Variables.

Mathematical formula of Belief Network:

the probabilities in the Belief Network are calculated by using following formula.

$$P(X_1, X_2, \ldots X_N) = \prod_{i=1}^{N} P\left(\frac{X_i}{Parents(X_i)}\right)$$

In order to calculate the joint "distribution" we need to have Conditional probability.

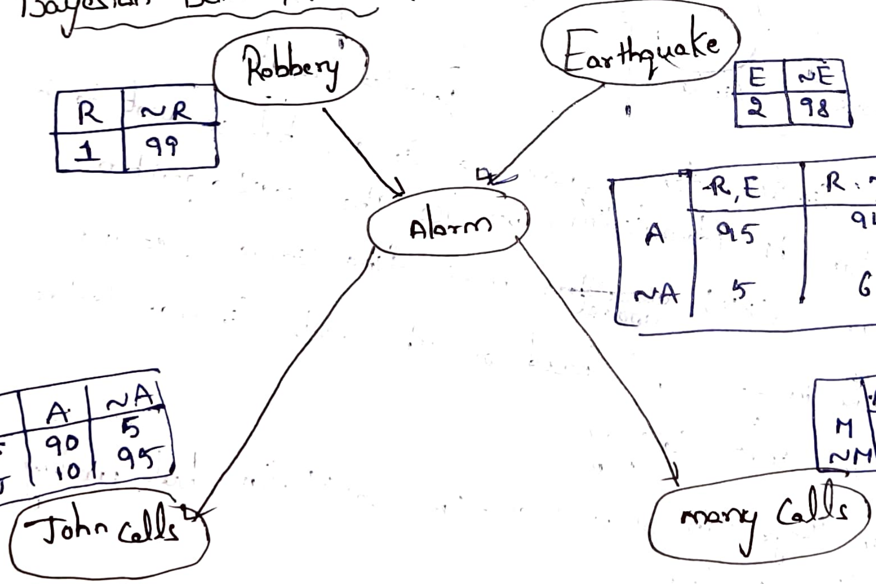
$P(J, M, A, \sim R, \sim E)$?

$P\left(\frac{J}{A}\right) \times P\left(\frac{M}{A}\right) \cdot P\left(\frac{A}{\sim R \sim E}\right) \cdot P(\sim R) \times P(\sim E)$

$90 \times 70 \times 1 \times 99 \times 98$

$= 56{,}133{,}000$

Bayesian Belief Network:



R	~R
1	99

E	~E
2	98

	R, E	R, ~E	~R E	~R ~E
A	95	94	29	1
~A	5	6	71	99

	A	~A
J	90	5
~J	10	95

	A	~A
M	70	1
~M	30	99

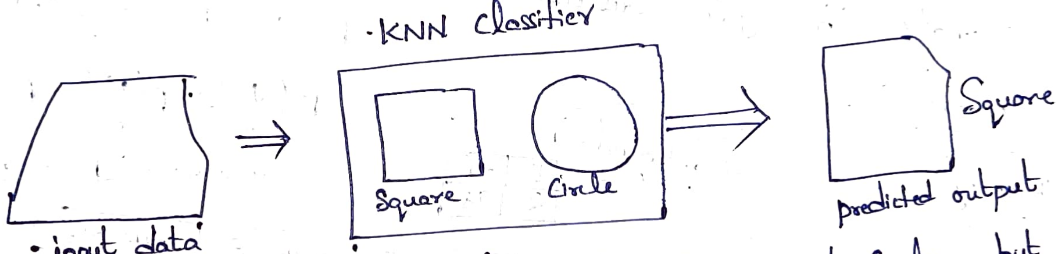
K - Nearest Neighbor (KNN) Algorithm :-

⇒ K-Nearest Neighbor is one of the Simplest machine learning algorithms based on Supervised Learning technique.

↳ in Supervised learning we train machines using labeled data (data which contains input & output)

⇒ KNN algorithm assumes the Similarity between the new data and available data and put the new data into the category that is most Similar to the available Categories.

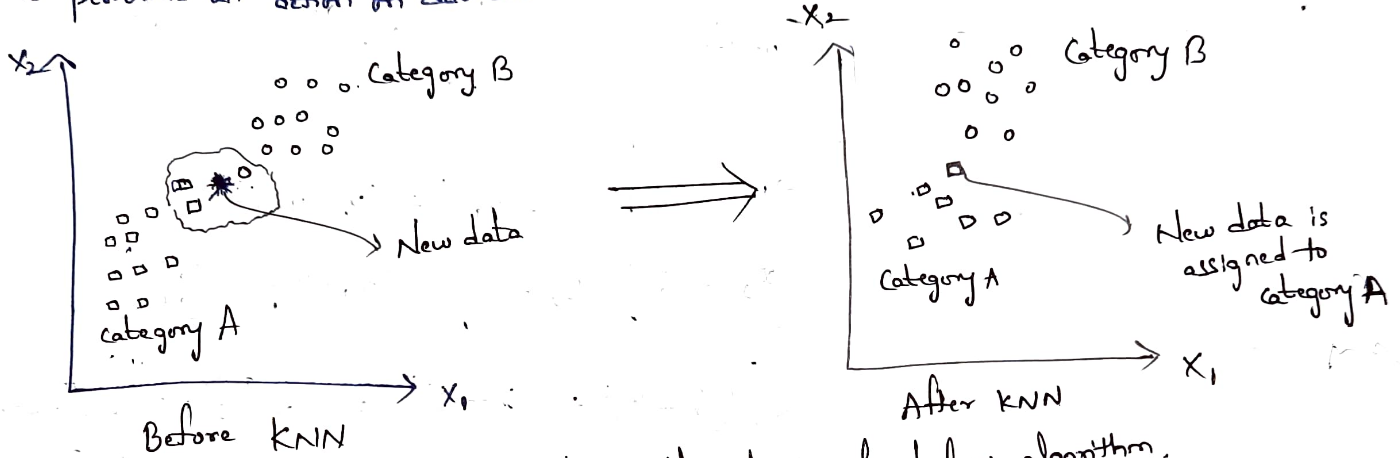
Example:



Suppose, we have an image that looks Similar to Square and Circle, but we want to know either it is a Square or Circle. So for this identification, we use the KNN algorithm, as it works on Similarity measure. Our KNN ~~algorithm~~ model will find Similar features of new data to Square and Circle images and based on the most Similar features it will put it in either Square or Circle category

⇒ KNN algorithm Can be used for Regression as well as for classification but mostly it is used for the classification problems.

⇒ It is also called as lazy learner algorithm because it does not learn from the training set immediately instead it stores the data set and at the time of classification it performs an action on dataset.



the KNN working can be explained on the basis of below algorithm.

Step 1:- Select the K number of neighbors

Step 2:- Calculate the Euclidean distance of K number of neighbors

Step 3:- Take the K ~~neighbors as~~ nearest neighbors as per the Calculated Euclidean distance.

Step 4:- Among these K neighbors, Count the number of data points in each Category

Step 5:- Assign the new data points to that Category for which the number of neighbor is maximum

Step 6:- our model is ready.

Example: perform KNN classification Algorithm on following dataset and predict the class for [maths = 6 , Computers = 8] where $k = 3$

	maths	Computers	Result
d_1	4	3	F
d_2	6	7	P
d_3	7	8	P
d_4	5	5	F
d_5	8	8	P

Euclidean distance $(d) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$d_1 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$\boxed{x_2 = 6}$ $\boxed{y_2 = 8}$ $\boxed{x_1 = 4}$ $\boxed{y_1 = 3}$

$$\sqrt{(6-4)^2 + (8-3)^2} = \sqrt{2^2 + 5^2} = \sqrt{4+25} = \sqrt{29}$$

5.38

$x_1 = 6, \ y_1 = 7$

$$d_2 = \sqrt{(6-6)^2 + (8-7)^2} = \sqrt{0+1} = 1 \checkmark$$

$$d_3 = \sqrt{(6-7)^2 + (8-8)^2} = \sqrt{1+0} = 1 \checkmark$$

$$d_4 = \sqrt{(6-5)^2 + (8-5)^2} = \sqrt{1+9} = \sqrt{10} = 3.16$$

$$d_5 = \sqrt{(6-8)^2 + (8-8)^2} = \sqrt{4+0} = \sqrt{4} = 2 \checkmark$$

	maths	Computers	R
d_2	6	7	P
d_3	7	8	P
d_5	8	8	P

for maths = 6, Computers = 8
Result is "P"

Naive Bayes Classifier Algorithm :-

train machines using labeled data

⇒ Naive Bayes algorithm is a Supervised learning algorithm, which is based on Bayes theorem and used for Solving classification problems

⇒ It is mainly used in text classification that includes a high-dimensional training dataset.

⇒ Naive Bayes classifier is one of the Simple and most effective classification algorithms which helps in building the fast machine learning models that can make quick predictions.

⇒ It is a probabilistic classifier, which means it predicts on the basis of the probability of an object.

Advantages: ① It is one of the fast and easy ML algorithms to predict a class of datasets

② It is the most popular choice for text classification problems

~~Disadvantages~~: The formula for Bayes theorem is given as

$$\boxed{P\left(\frac{A}{B}\right) = \frac{P(B/A) \cdot P(A)}{P(B)}}$$

probability of B when A is true.

Probability of A when B is true

Example: From the Given dataset, Apply Naive-Baye's Algorithm and predict that if a fruit has the following properties then which type of fruit it is?

Fruit = {yellow, Sweet, Long} X

dataset

Fruit	yellow	Sweet	long	Total
mango	35	45	0	65
Banana	40	30	35	40
orange	5	10	5	15
total	80	85	40	120

Formula:

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

$P(\text{mango}) = 0$,,

$$P\left(\frac{X}{\text{mango}}\right) = P\left(\frac{\text{yellow}}{\text{mango}}\right) \times P\left(\frac{\text{Sweet}}{\text{mango}}\right) \times P\left(\frac{\text{long}}{\text{mango}}\right)$$

$$\frac{P\left(\frac{\text{mango}}{\text{yellow}}\right) \times P(\text{yellow})}{P(\text{mango})} \times \frac{P\left(\frac{\text{mango}}{\text{Sweet}}\right) \times P(\text{Sweet})}{P(\text{mango})} \times \frac{P\left(\frac{\text{mango}}{\text{long}}\right) \times P(\text{long})}{P(\text{mango})}$$

$$\frac{\frac{35}{80} \times \frac{80}{120}}{\frac{65}{120}} \times \frac{\frac{45}{85} \times \frac{85}{120}}{\frac{65}{120}} \times \frac{\frac{0}{40} \times \frac{40}{120}}{\frac{65}{120}}$$

$$= \frac{35}{\cancel{80}} \times \frac{\cancel{80}}{\cancel{120}} \times \frac{\cancel{120}}{65} \times \frac{45}{\cancel{85}} \times \frac{\cancel{85}}{\cancel{120}} \times \frac{\cancel{120}}{65} \times \frac{0}{\cancel{40}} \times \frac{\cancel{40}}{\cancel{120}} \times \frac{\cancel{120}}{65}$$

$$= \frac{35}{65} \times \frac{45}{65} \times \frac{0}{65}$$

$$= 0$$,,

$$P(\text{Banana}) \implies P\left(\frac{X}{\text{Banana}}\right) = P\left(\frac{\text{yellow}}{\text{Banana}}\right) \times P\left(\frac{\text{Sweet}}{\text{Banana}}\right) \times P\left(\frac{\text{long}}{\text{Banana}}\right)$$

(0.65)

$$= \frac{P\left(\frac{\text{Banana}}{\text{yellow}}\right) \times P(\text{yellow})}{P(\text{Banana})} \times \frac{P\left(\frac{\text{Banana}}{\text{Sweet}}\right) \times P(\text{sweet})}{P(\text{Banana})} \times \frac{P\left(\frac{\text{Banana}}{\text{long}}\right) \times P(\text{long})}{P(\text{Banana})}$$

$$= \frac{\frac{40}{80} \times \frac{80}{120}}{40/120} \times \frac{\frac{30}{85} \times \frac{80}{120}}{40/120} \times \frac{\frac{35}{40} \times \frac{40}{120}}{\frac{40}{120}}$$

$$= \frac{\cancel{40}}{\cancel{80}} \times \frac{\cancel{80}}{\cancel{120}} \times \frac{\cancel{120}}{40} \times \frac{30}{\cancel{85}} \times \frac{\cancel{85}}{\cancel{120}} \times \frac{120}{40} \times \frac{35}{\cancel{40}} \times \frac{\cancel{40}}{\cancel{120}} \times \frac{\cancel{120}}{40}$$

$$= 1 \times \frac{30}{40} \times \frac{35}{40}$$

$$= 0.65$$

$$P(\text{orange}) = P\left(\frac{\text{yellow}}{\text{orange}}\right) \times P\left(\frac{\text{Sweet}}{\text{orange}}\right) \times P\left(\frac{\text{long}}{\text{orange}}\right)$$

0.07

$$= \frac{P\left(\frac{\text{orange}}{\text{yellow}}\right) \times P(\text{yellow})}{P(\text{orange})} \times \frac{P\left(\frac{\text{orange}}{\text{Sweet}}\right) \times P(\text{Sweet})}{P(\text{orange})} \times \frac{P\left(\frac{\text{orange}}{\text{long}}\right) P(\text{long})}{P(\text{orange})}$$

$$= \frac{\frac{5}{80} \times \frac{80}{120}}{\frac{15}{120}} \times \frac{\frac{10}{85} \times \frac{85}{120}}{\frac{15}{120}} \times \frac{\frac{5}{40} \times \frac{40}{120}}{\frac{15}{120}}$$

$$= \frac{5}{80} \times \frac{80}{120} \times \frac{120}{15} \times \frac{10}{85} \times \frac{85}{120} \times \frac{120}{15} \times \frac{5}{40} \times \frac{40}{120} \times \frac{120}{15}$$

$$= \frac{5}{15} \times \frac{10}{15} \times \frac{5}{15}$$

$$\left\{ \begin{array}{l} P(B) = 0.65 \\ P(m) = 0 \\ P(O) = 0.07 \end{array} \right\}$$

$$= 0.07$$

$\therefore$ fruit = {yellow, Sweet, long} belongs to Banana data gain

Decision Tree :-

⇒ Decision tree algorithm falls under the Category of Supervised learning. they can be used to Solve both regression and classification problems. It is a tree that helps us in decision making purposes. the decision tree Creates classification or regression models as a tree structure. ie, Decision tree uses the tree Representation to Solve the problem

Decision Tree Contains 3 types of nodes, they are

1. root node: It is the top most node in the tree. data which is inside the node is Known as attribute.
2. Internal node: each internal node denotes a test on attribute. nodes which are in between root node and leaf nodes, are called as internal nodes.
3. leaf node: we call last nodes as leaf nodes. they represents output ie, class label

⇒ root node and Internal nodes are represented by Rectange and leaf nodes are represented by ovals

Advantages:

1. It does not require any domain Knowledge
2. Classification Steps of decision tree are Simple and fast.
3. missing values in data does not effect output.
4. A decision tree model is automatic and does not require a Standardization of data.

checking data is in correct format or not

Key Factors: Building a decision tree is all about discovering attributes that return the highest data gain 8

Entropy:

Entropy refers to a common way to measure impurity. In the decision tree, it measures the impurity in data set.

Information Gain:

Information Gain refers to the decline in entropy after the dataset is split. It is also called as Entropy Reduction.

Impure Attribute

Name	Gender	Salary
A	M	50K
B	F	30K
C	F	10K
D	M	60K
E	F	10K

Construct decision tree for salary ~~50K and~~ people with salary

Name	Gender
A	80k
B	30k
C	10k
D	60k
E	10k

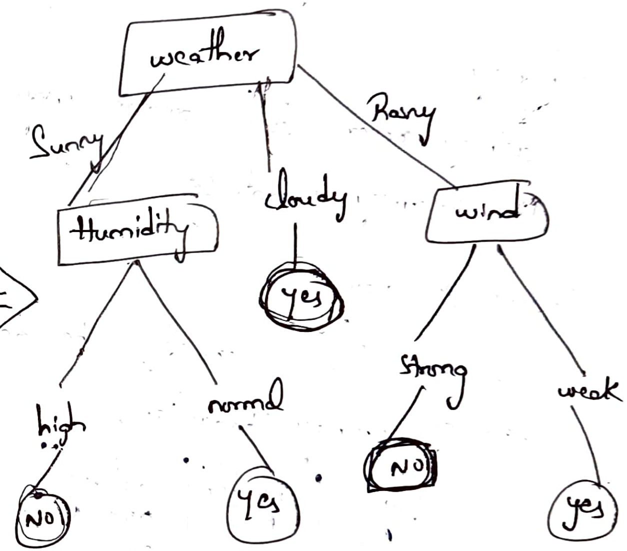
Information Gain

Gender

Example:

Day	weather	Temperature	Humidity	wind	play ?
1	Sunny	Hot	high	weak	No
2	cloudy	Hot	high	weak	yes
3	cloudy	mild	high	strong	yes
4	Rainy	mild	High	Strong	No
5	Sunny	mild	Normal	Strong	yes
6	Rainy	Cool	Normal	Strong	No
7	Rainy	mild	high	weak	yes
8	Sunny	Hot	high	Strong	No
9	cloudy	Hot	Normal	weak	yes
10	Rainy	mild	high	Strong	No

$\Longrightarrow$



Rules: 1) if weather = cloudy, then play = yes

2) if weather = Sunny, humidity = high, then play = No

3) " " " = normal " play = yes

4) if weather = Rainy, wind = strong then play = No

5) " " " = weak then play = yes

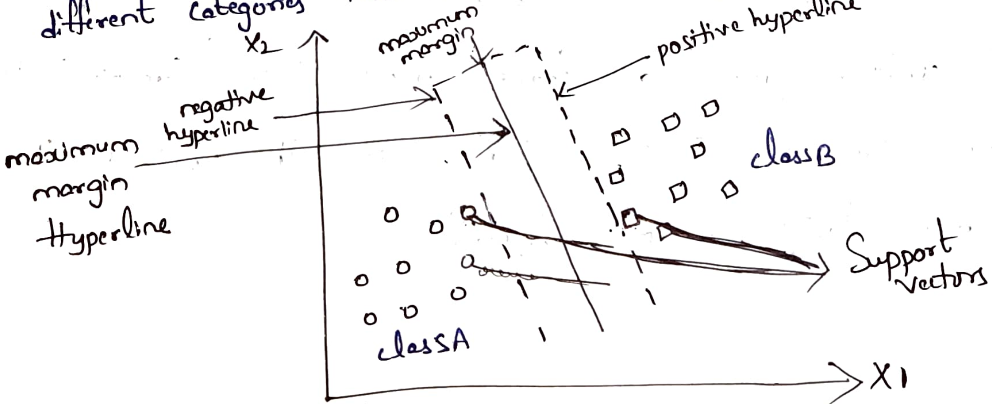
day = 11, weather = cloudy, Tem = Hot,
Humidity = high, wind = weak,
play = ?
= yes

Support Vector Machine (SVM):

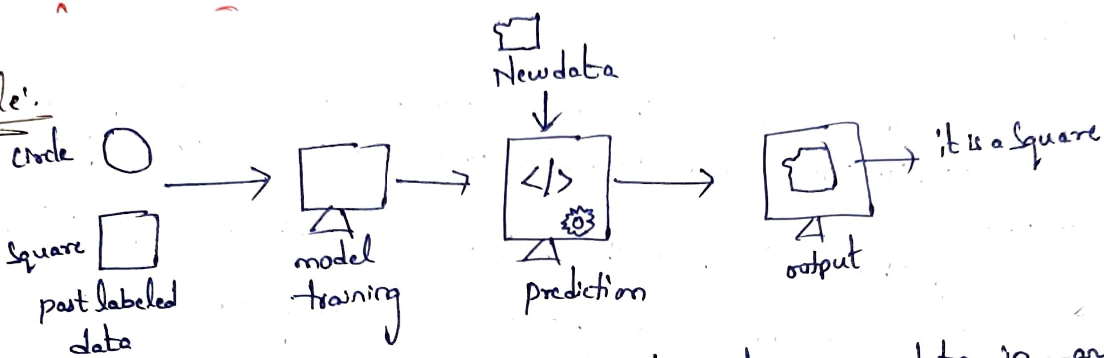
⇒ Support Vector Machine or SVM is one of the most popular Supervised learning algorithms, which is used for classification as well as Regression problems. However, primarily, it is used for classification problems.

⇒ The goal of the SVM algorithm is to create the best line or decision boundary that can separate n-dimensional space into classes so that we can easily put new data point in the correct category in the future. We call this best line or decision boundary as hyperline.

⇒ SVM chooses the extreme points/vectors that help in creating the hyperline. These extreme cases are called as Support vectors, and hence algorithm is termed as Support Vector machine. Consider the below diagram in which there are two different categories that are classified using a decision boundary or hyperline.



Example:



Suppose we have new data, i want to place this new data in any of the category ie either in Circle or Square Category. So we want a model that can accurately identify whether it is a Circle or Square. So Such a model Can be Created by using the SVM algorithm. we will first train our model with lots of images of Square and Circle so that it Can learn about different features of Circle and Square. So as Support Vector Creates a decision boundary between these two data (circle & square) and choose extreme Cases (Support Vectors) it will see the extreme Case of Circle and Square. on the basis of the Support Vectors, it will classify it as Square.

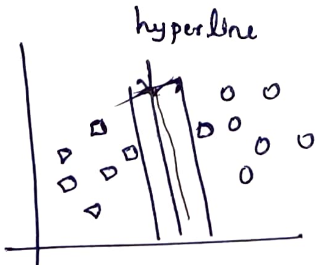
Types of SVM:-

SVM can be of two types:

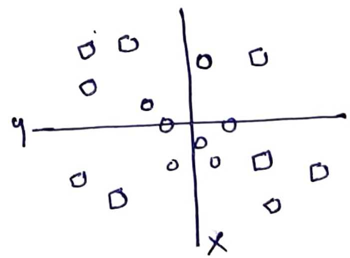
① **Linear SVM:-** linear SVM is used for linearly Separable data, which means if a dataset can be classified into two classes by using a Single Stright line, then Such a data is termed as linearly Separable data, and classifier used is called as Linear SVM classifier.

② **Non-linear SVM:-**

Non-Linear SVM is used for non-linearly Separated data, which means if a data set cannot be classified by using a Stright line, then Such data is termed as non linear data and classifiers used is called as Non-linear SVM classifier.



Linear SVM



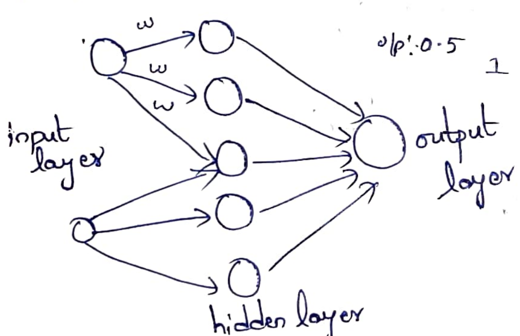
Non linear SVM.

Classification By Back propagation:-

⇒ Back propagation is an algorithm that back propagates the errors from output nodes to the input nodes. therefore, it is Simply referred to as backward propagation of errors. It is used in Various applications of neural networks in data mining like character recognition, Signature Verification etc.

⇒ **Neural Network:-**

Artificial neural networks, usually Simply called neural networks are Computing Systems with interconnected nodes that work much like neurons in human brain.



A Simple neural network

working of Back propagation:-

⇒ Neural network use Supervised learning to generate output vectors from input vectors in the network. It Compares generated output to desired output and generates an error report if the result doesnot match the generated output vector. then it adjusts the weights according to the error report to get desired output.

Need for Backpropagation:-

Backpropagation is "Back propagation of errors" and is very useful for training neural networks. It's fast, easy to implement, and Simple. Backpropagation doesnot require any parameters to be set, except the number of inputs. Backpropagation is ~~flexib~~ flexible method because no prior Knowledge of network is required.

Advantages:-

1. It is Simple, fast and easy to program
2. only numbers of input are required, no need of any other parameters.
3. It's flexible and efficient.
4. No need for users to learn any Special functions

Disadvantages:-

1. Noisy data leads to inaccurate Results
2. performance is highly dependent on input data
3. Spending too much time training

Types of Back propagation:

There are two types of back propagation networks

1. Static backpropagation:

Static Backpropagation is a network designed to map static inputs for static outputs. These types of networks are capable for solving static classification problems such as optical character Recognization

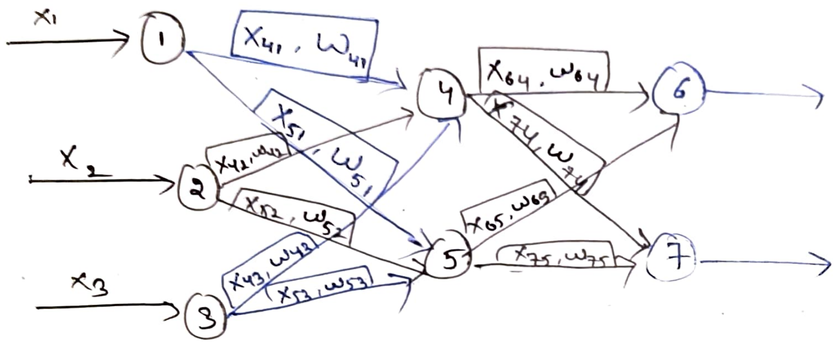
2. Recurrent back propagation:

It is another network used for fixed point learning. Activation in recurrent backpropagation is feed forward until a fixed value is reached.

⇒ the key differences is Static back propagation offers immediate mapping, while mapping recurrent backpropagation is not immediate

⇒ Back propagation contains network inputs (n_{in}). hidden layer (n_{hidden}) and output unit (n_{out})

⇒ the input from unit i to unit j is denoted by X_{ji} and weights from unit i to unit j is denoted by w_{ji}



Back propagation Algorithm:

Step 1: Inputs x, arrived through the pre connected path.

Step 2: The input is modeled using true weights w. weights are usually choosen randomly.

Step 3: calculate the output of each neuron from the input layer to the hidden layer to the output layer

Step 4: calculate the error in outputs

Step 5: From the output layer, go back to the hidden layer to adjust the weights to reduce errors

Step 6: Repeat the process until the desired output is achieved.